

Learn2Push: Learning Non-Prehensile Manipulation Across Object Families with Legged Mobile Manipulators

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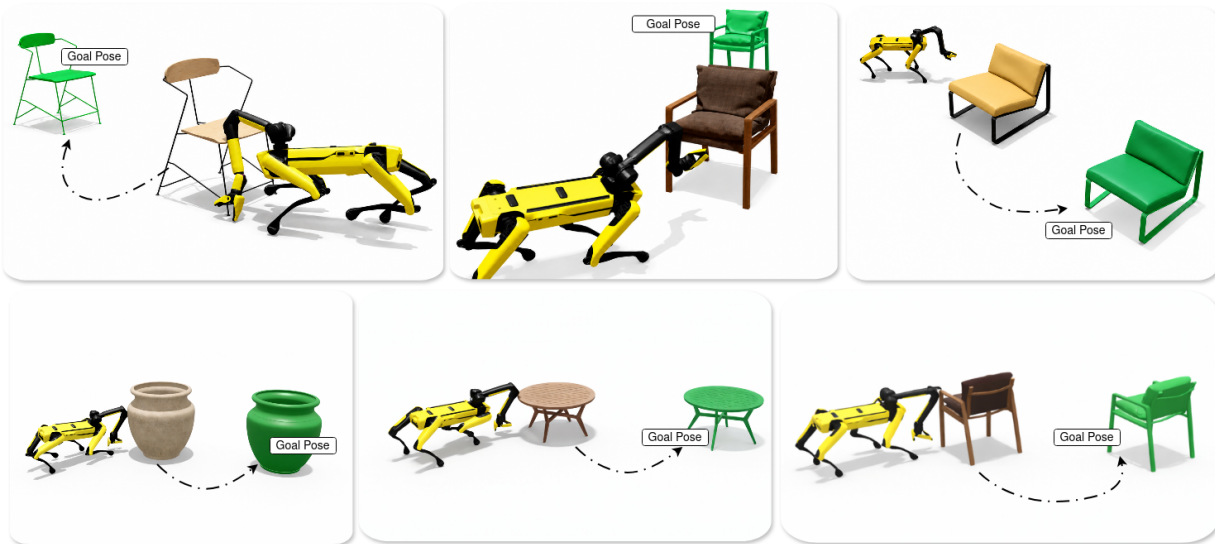


Fig. 1. **Arm-mediated pushing across object geometries.** The learned policy coordinates the legged base and manipulator to push chairs, a planter, and a table toward target poses (green), illustrating the range of object geometries considered in this work.

Abstract—Non-prehensile manipulation enables robots to reposition objects through contact across diverse geometries and physical properties without relying on grasping. We study goal-directed arm-mediated object pushing with a quadrupedal mobile manipulator and propose a hierarchical reinforcement learning framework that decouples task-level decision making from low-level dynamic stabilization. The high-level policy outputs target base velocities and arm-joint targets. A pre-trained low-level locomotion policy tracks the base command to produce dynamically stable locomotion, while a separate arm controller tracks the commanded joint targets. To guide exploration toward feasible robot–object interactions, we introduce geometry-guided contact sampling from the object point cloud and use the sampled target to shape end-effector reaching before shifting the emphasis toward the object-pose objective. We evaluate performance and robustness across object families and distribution shifts. The learned policy remains effective under geometry and physics variation and generalizes to unseen everyday-object geometries. Ablations further show that contact guidance, the staged reward curriculum, and online adaptation are important for learning and transfer.

I. INTRODUCTION

Robots in human environments often need to move objects that are bulky, heavy, or inconvenient to grasp—for example, chairs, carts, bins, and planters. Non-prehensile pushing avoids grasp acquisition and can exploit a mobile robot’s workspace, but it introduces sustained, intermittent

contact whose outcome depends strongly on object geometry, friction, mass, and contact location [1], [2], [3]. For a legged mobile manipulator, the policy must additionally coordinate the arm with a dynamically stabilized base.

Recent learning systems have demonstrated object pushing with legged robots [4], [5], [6] and hierarchical locomotion–manipulation controllers [7], [8]. These systems motivate separating task-level interaction from locomotion stabilization, but contact acquisition remains difficult for geometrically varied objects. A target sampled from one hand-selected plane can guide interaction with a box, yet the same assumption is brittle for chairs and other non-convex meshes. Our earlier work studied whole-body quadrupedal pushing under geometry and physics variation [9]; here, we isolate arm-mediated contact acquisition and extend evaluation across non-convex object families.

We focus exclusively on *arm-mediated* (AM) pushing: the arm or end effector establishes the intended object contact, while the legs provide locomotion and balance. Our system couples a high-level goal-conditioned policy to a frozen locomotion prior and introduces point-cloud contact sampling to make useful interaction discoverable during reinforcement learning. The paper makes two contributions: (1) a geometry-guided sampling and reward curriculum that converts arbitrary object meshes into reachable contact intents without object-specific contact annotations; and (2) an AM pushing study spanning randomized parametric objects,

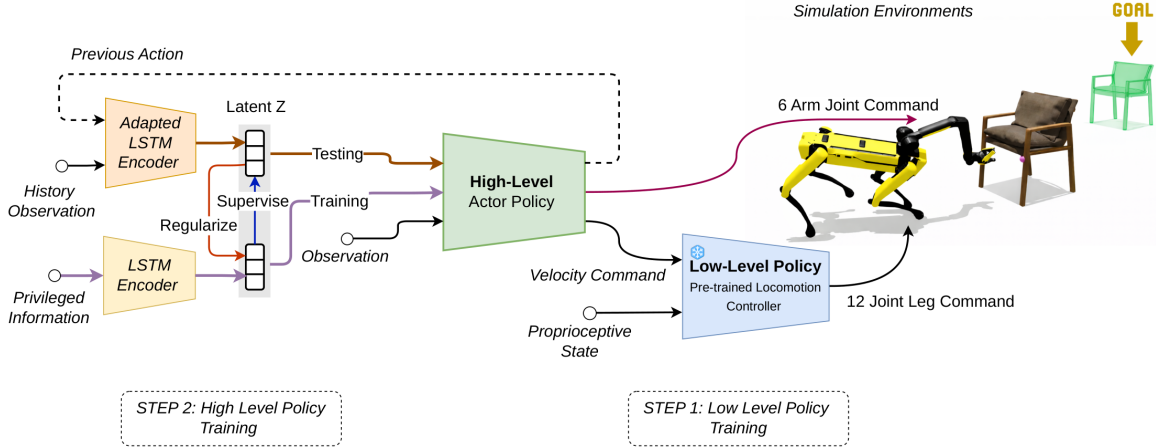


Fig. 2. **Hierarchical arm-mediated pushing.** The high-level actor receives task observations and an adaptation latent, then jointly outputs planar base velocity and six arm-joint targets. A frozen locomotion policy converts the base command and proprioception into leg targets; arm targets are tracked by joint controllers. During training, a privileged encoder supervises the history encoder used at test time.

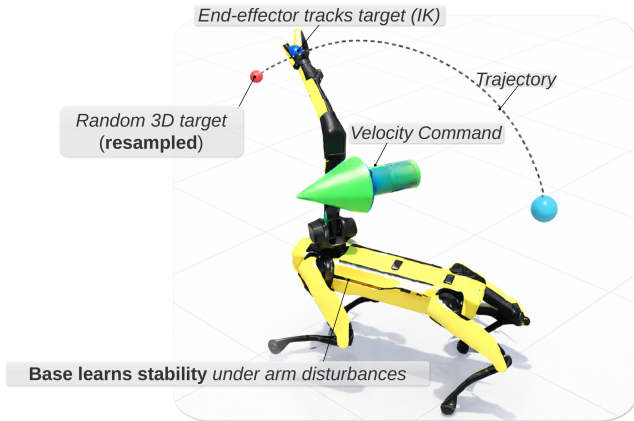


Fig. 3. **Locomotion-prior pretraining.** The arm tracks resampled 3-D targets while the base follows velocity commands, training leg control to remain stable across arm configurations and disturbances.

shifted geometry and physics, unseen everyday objects, and cross-chair ablations.¹

II. RELATED WORK

Non-prehensile pushing. Classical pushing work characterizes the relationship between contact, friction, and planar object motion [2], [3]. Learning-based methods reduce dependence on exact mechanics and have demonstrated simulation-to-physical transfer for tabletop non-prehensile manipulation [10]. Mobile manipulation extends the problem because the robot must continually reposition its base while regulating object contact. Geometry-aware mobile-manipulation methods have also used compact shape abstractions to estimate grasp poses on large objects [11]; in contrast, we sample reachable mesh surfaces to acquire non-prehensile contacts without requiring a grasp. Prior systems plan feedback-controlled pushes for carts and objects with

passive casters [12], [13], while recent RL approaches learn goal-directed pushing with mobile manipulators [5]. Our focus differs in using a legged platform and in learning contact acquisition directly from general object geometry.

Legged loco-manipulation. Hierarchical policies have enabled quadrupeds to manipulate objects while retaining robust locomotion [4], [7], [8]. Visual policies and large-scale randomized training further improve generalization [6], [14]. We adopt the established separation between a task policy and a locomotion prior; we do not claim this hierarchy as a new learning primitive. Instead, we study how geometry-derived contact intents, staged rewards, and online adaptation make arm-mediated pushing learnable across object families.

III. APPROACH

A. Task and Hierarchical Controller

Given an initial object pose (p_o, ψ_o) and a planar goal pose (p_g, ψ_g) , the objective is to bring the object to the goal without toppling it or making unintended contact with the robot body. The high-level policy acts through

$$a_t = [v_x, v_y, \omega_z, q_{1:6}^{\text{arm}}] \in \mathbb{R}^9, \quad (1)$$

where the first three entries are body-frame planar velocity commands and the remaining entries are arm-joint targets; the gripper remains fixed. The locomotion policy is pre-trained while the arm tracks randomly sampled end-effector targets, exposing the base to a broad range of upper-body disturbances. It is then frozen and maps the commanded base velocity and proprioception to 12 leg-joint targets. This hierarchy isolates gait stabilization from contact-rich task exploration (Fig. 2).

The actor observes base linear and angular velocity, projected gravity, previous action, arm positions and velocities, end-effector-to-object displacement, object-to-goal displacement, and relative object/goal rotations. To adapt to latent dynamics, we follow regularized online adaptation [15]: a teacher LSTM encodes a 24-step history of privileged

¹Supplementary: <https://learn2push.github.io>

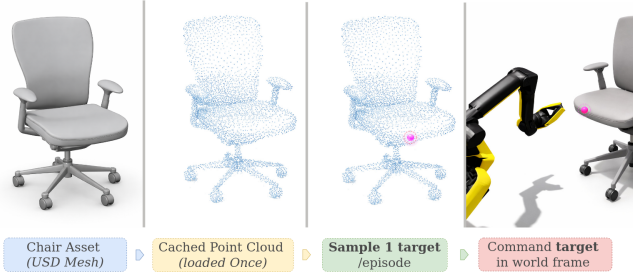


Fig. 4. **Geometry-guided contact sampling.** An object mesh is converted to a cached surface cloud, filtered to retain reachable interaction regions, and sampled once per episode. The selected contact is transformed to the world frame to guide end-effector approach.

physical parameters into $z_\psi \in \mathbb{R}^{16}$, while a student LSTM predicts z_ϕ from observation–action history. The encoders are trained using

$$L_{\text{adapt}} = \lambda \|z_\phi - \text{sg}(z_\psi)\|_2^2 + \|\text{sg}(z_\phi) - z_\psi\|_2^2, \quad (2)$$

and only the student is used at test time. The actor is optimized with PPO [16].

B. Geometry-Guided Contact Acquisition

At the start of an episode, the policy must approach a useful surface before object progress produces a meaningful learning signal. We therefore cache a dense point cloud from each object’s mesh and construct an interaction set $\mathcal{S}_{\text{int}}$ by removing points that are interior, too close to the ground, outside the manipulator’s reachable region, or located on locally irregular surface patches. One point $p_c \sim \mathcal{U}(\mathcal{S}_{\text{int}})$ is sampled per episode and transformed into the world frame (Fig. 4). This sampling does not prescribe a full pushing trajectory; it provides an initial contact intent while leaving the policy free to reposition and regulate contact.

We combine contact acquisition with pose progress and safety:

$$r_{\text{reach}} = \exp(-\|p_{ee} - p_c\|_2^2 / \sigma_r^2), \quad (3)$$

$$r_{\text{pose}} = \exp(-\|K_o - K_g\|_2^2 / \sigma_p^2), \quad (4)$$

where K_o and K_g are corresponding object and goal key-points. Additional terms reward object velocity toward the goal and smooth actions, while penalizing tilt and contacts outside the arm/end effector. Early training weights reaching strongly ($w_{\text{reach}} = 2.5$). After 2000 iterations this weight is reduced to 0.63, and after 4000 iterations the sparse pose-completion weight is increased to 4.0. The curriculum first teaches contact acquisition, then shifts control authority toward object transport and final pose regulation.

IV. EXPERIMENTS

We train in Isaac Lab [17] with 4096 parallel environments for 20,000 iterations. Each 30 s episode randomizes the robot, object, and goal in an 8 m $\times$ 8 m arena. Table II summarizes the train-time randomization and the expanded geometry and physics ranges used for OOD evaluation.

An episode succeeds when the object remains stationary within the pose tolerance for four consecutive high-level

TABLE I
AM TASK REWARD. COMPLETION AND REACHING USE THE CURRICULUM DESCRIBED IN THE TEXT.

Term	Form	w	σ
Pose alignment	$\exp(-\ K_o - K_g\ _2^2 / \sigma^2)$	4.00	0.78
Reach contact	$\exp(-\ p_{ee} - p_c\ _2^2 / \sigma^2)$	2.50 $\rightarrow$ 0.63	0.60
Velocity to goal	$\exp((\hat{v}_o^T \hat{d}_g - 1) / \sigma^2)$	0.16	0.71
Object stability	$[1 + \exp(500(\theta - 20^\circ))]^{-1}$	0.40	–
Sparse completion	$\mathbf{1}(e_{xy} \leq 5 \text{ cm} \wedge e_\psi \leq 5^\circ)$	0.10 $\rightarrow$ 4.00	–
Action rate	$\exp(-\ a_t - a_{t-1}\ _2^2 / \sigma^2)$	0.30	0.35
Action acceleration	$\exp(-\ a_t - 2a_{t-1} + a_{t-2}\ _2^2 / \sigma^2)$	0.30	0.38

TABLE II
TRAIN AND OOD RANDOMIZATION RANGES.

Factor	Train	OOD
Mass (kg)	[2, 8]	[2, 18]
Static friction	[0.6, 1.2]	[0.1, 0.6]
Dynamic friction	[0.48, 0.96]	[0.08, 0.48]
CoM offset (cm)	≤ 5	≤ 20
Width/length (m)	[0.25, 0.75]	[0.25, 1.25]
Height (m)	[0.4, 1.0]	[0.4, 1.8]
Goal distance (m)	[1.5, 2.5]	[0.8, 3.5]

steps. We report C1 (15 cm, 15°) and the stricter C2 (10 cm, 10°), plus terminal position and yaw errors. An episode terminates if the object tilts beyond 20°, contacts the ground, or the robot violates the intended-contact constraints.

A. Generalization and Contact Behavior

Table III shows high ID success for both parametric families. Under novel dimensions and aspect ratios, C1 remains 74.4–76.2%, indicating that the learned policy tolerates substantial changes in accessible contact geometry. The physics shift is harder: low friction and heavier objects increase residual error, particularly cylinder translation, but C1 remains 66.9–70.0%.

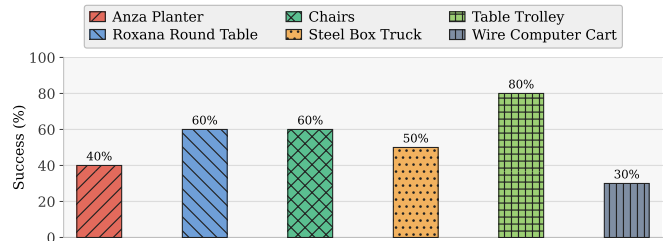


Fig. 5. **Success on unseen object geometries.** Category-level AM success on seven chairs (reported as their mean) and five individual non-chair objects.

We also evaluate the trained policy on 12 unseen rigid meshes: seven chairs and five non-chair everyday objects (Fig. 5). Mean chair success is 60%; among individual non-chair objects, success ranges from 30% on the wire computer cart to 80% on the table trolley. The round table and steel box truck reach 60% and 50%, respectively, while the smooth Anza planter reaches 40%. Failures concentrate on mechanics outside training: casters roll instead of sticking, while smooth, near-axisymmetric surfaces produce changing contact normals and ambiguous yaw.

TABLE III
AM PERFORMANCE OVER RANDOMIZED PARAMETRIC OBJECTS.
EACH SUITE CONTAINS 500 VARIANTS PER REPORTED FAMILY.

Suite	Object	C1 (%)	C2 (%)	e_{pos} (cm)	e_{ψ} (deg)
ID	Cuboid	91.88	85.63	8.07	3.85
	Cylinder	92.46	86.35	7.73	3.88
OOD geometry	Cuboid	76.24	61.80	13.58	9.98
	Cylinder	74.37	65.00	11.94	15.18
OOD physics	Cuboid	66.87	51.88	14.34	10.39
	Cylinder	70.00	66.87	24.86	14.71

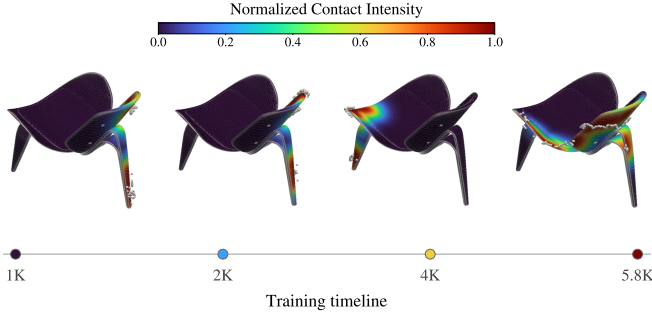


Fig. 6. **Interaction footprint over training.** Projected contacts progress from diffuse exploration to repeatable regions; color denotes normalized kernelized intensity.

TABLE IV
CROSS-CHAIR ABLATION. TRAIN ON ONE CHAIR; EVALUATE ON UNSEEN CHAIRS.

Variant	C1 (%)	C2 (%)	e_{pos} (cm)	e_{ψ} (deg)
No sampled reach reward	0.00	0.00	—	—
No sparse-reward curriculum	6.25	0.00	184.46	77.89
No adaptation encoder	39.00	36.91	51.19	44.17
Learn2Push	69.03	67.64	29.70	18.75
Privileged-input expert	79.86	69.51	12.37	13.10

Contact projections also reveal how interaction changes during learning. Early contacts are diffuse across the chair; later checkpoints concentrate on accessible, mechanically useful regions (Fig. 6). This behavior is emergent: sampled targets guide exploration, but the policy is not given a labeled “best” pushing face. Together, these results show transfer across rigid geometry while exposing unmodeled mobility as a remaining challenge.

B. Ablations and Failure Analysis

We train on one chair and test on held-out chair geometries (Table IV). Without the sampled reach reward, the policy never discovers a successful behavior. Without the sparse-reward curriculum, it exploits easier base/leg contacts and fails under the strict criterion. Removing the adaptation encoder also reduces C2 from 67.64% to 36.91%, consistent with history helping infer geometry- and dynamics-dependent interaction responses. The privileged-input expert indicates remaining headroom rather than a deployable policy.

Aggregate success rates obscure which component of precise placement fails. We therefore partition trials that

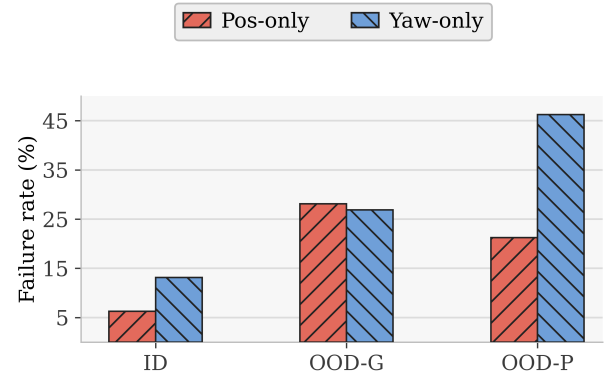


Fig. 7. **AM failure attribution under distribution shift.** Rates of trials that violate only the position or only the yaw component of C2. OOD physics is dominated by residual yaw error.

miss C2 into *position-only* failures, which satisfy the yaw tolerance but miss the position tolerance, and *yaw-only* failures, which satisfy position but miss yaw (Fig. 7). ID failures are infrequent and are more often yaw-limited. Under OOD geometry, the two failure types occur at comparable rates, consistent with new surface normals affecting both translation and rotation. OOD physics produces a clearer asymmetry: yaw-only failures dominate, suggesting that altered friction, mass, and center of mass primarily disrupt final orientation regulation after the object has reached the goal neighborhood. This diagnosis motivates orientation-aware corrective contacts rather than simply increasing reach shaping.

C. Future Work

The failure analysis points to three immediate extensions: online contact re-sampling for orientation correction, mobility-aware state and dynamics for rolling objects, and closed-loop hardware evaluation with onboard pose estimation and force-aware feedback. Physical trials should report perception, contact-acquisition, transport, and final-pose failures separately, while enforcing conservative force/torque limits and a monitored stop.

V. CONCLUSION

Learn2Push frames generalizable non-prehensile pushing as purposeful arm-contact acquisition followed by coordinated base-arm regulation. Across randomized and unseen objects, geometry-guided contact intents make sparse pushing learnable, while the curriculum and history encoder support transfer.

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